AP PRECALCULUS

Unit 1 • Lesson 1.14 practice

Lesson 1.14 • Function Model Construction and Application

4 multiple-choice questions and 1 free-response question

Question 1

A rectangle has perimeter 40 cm and width w cm. Which function gives its area?

(A) $A(w) = w(40 - w)$
(B) $A(w) = w(20 - w)$
(C) $A(w) = 2w + 40$
(D) $A(w) = w^2 - 20$

Question 2

What is a reasonable domain for $A(w)$ in the previous question?

(A) $0 < w < 20$
(B) $0 < w < 40$
(C) $w > 0$
(D) All real numbers

Question 3

What is the maximum area of a rectangle with perimeter 40 cm?

(A) 400 cm^2
(B) 200 cm^2
(C) 100 cm^2
(D) 80 cm^2

Question 4 (calculator)

An open box is made from a 12 in by 12 in square sheet by cutting squares of side x from the corners. What is the maximum volume?

(A) 96 in^3
(B) 108 in^3
(C) 144 in^3
(D) 128 in^3

Question 5 (Free response • calculator)

A cylindrical can must hold 500 cm^3. Let r be its radius and h its height, in centimeters.

(a) Express h in terms of r.
(b) Show that the surface area is $S(r) = 2\pi r^2 + \dfrac{1000}{r}$.
(c) Use a graphing calculator to find the radius that minimizes S, and the minimum surface area.
(d) Describe the behavior of S as $r \to 0^+$ and as $r \to \infty$, and explain why in context.